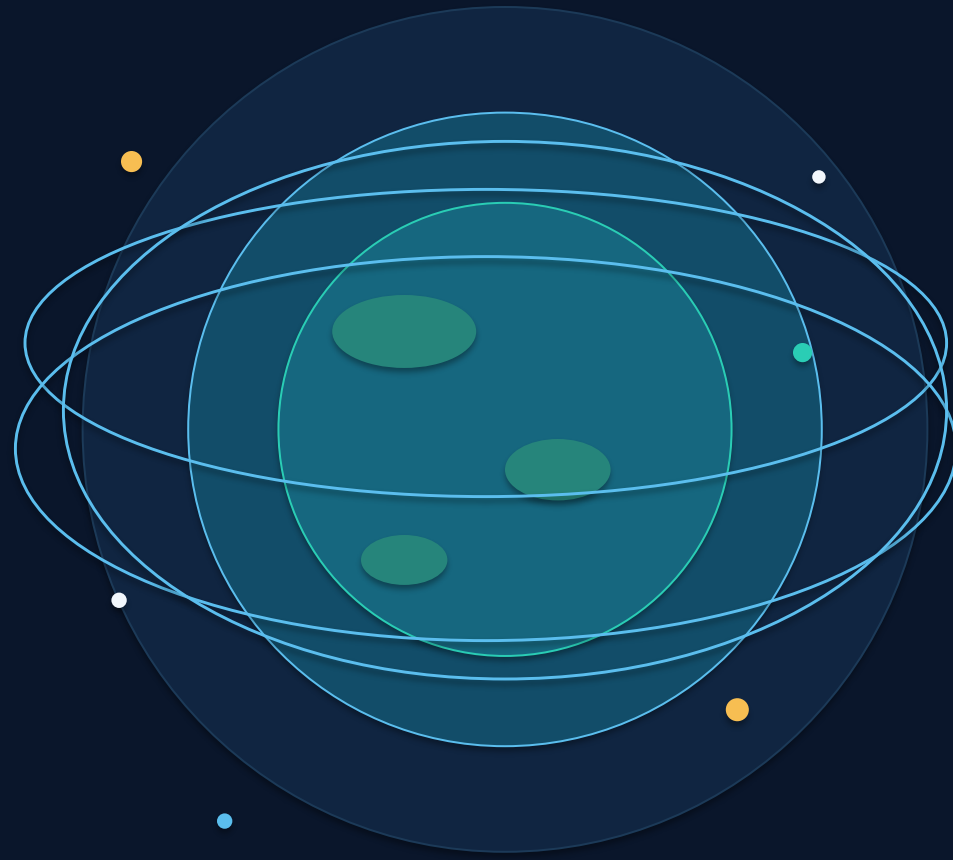
ORBITGUARD – AI-Assisted Space Debris Risk Prediction

THE PROBLEM: Thousands of satellites, rocket bodies and debris objects occupy Earth orbit. Their future positions are uncertain, making it difficult to identify which close approaches could become dangerous.

OUR GOAL: Predict future close approaches and re-entry risks, prioritize high-concern events, and explain the factors behind each risk alert.

DOMAIN: Defense, Aerospace, Navy, Government, Sovereign AI



TEAM NAME: CORE 5

Members:

1) Siddangouda Patil
2) Sudarshan Kulkarni
3) Vaishakhi Waddinavar
4) Vijita Patil
5) Mahi Kulkarni

TARGET USERS : Small-satellite operators • University satellite teams • Mission planners • SSA researchers

Proposed solution & objectives

A focused screening prototype that combines **physics-based orbital propagation** with **explainable risk prioritization**.

THE IDEA

Combine orbital mechanics with explainable risk scoring.

Propagate selected objects using SGP4, screen for close approaches, then rank events using transparent rules. Add ML only if suitable labelled data is available.

Important: public TLEs usually lack covariance. The prototype will show a risk score, not claim a calibrated collision probability unless adequate uncertainty data is available.

KEY OBJECTIVES

1	**Track**	Load and validate orbital elements for a manageable object set.
2	**Predict**	Propagate future positions and identify close approaches.
3	**Prioritize**	Rank events using separation, relative velocity, data freshness and uncertainty indicators.
4	**Explain**	Show the main factors behind each risk flag and provide clear alerts.
5	**Evaluate**	Benchmark predictions and measure errors, detection and false alerts.

Innovation & existing alternatives

Our Differentiator: Explainable Risk Prioritization

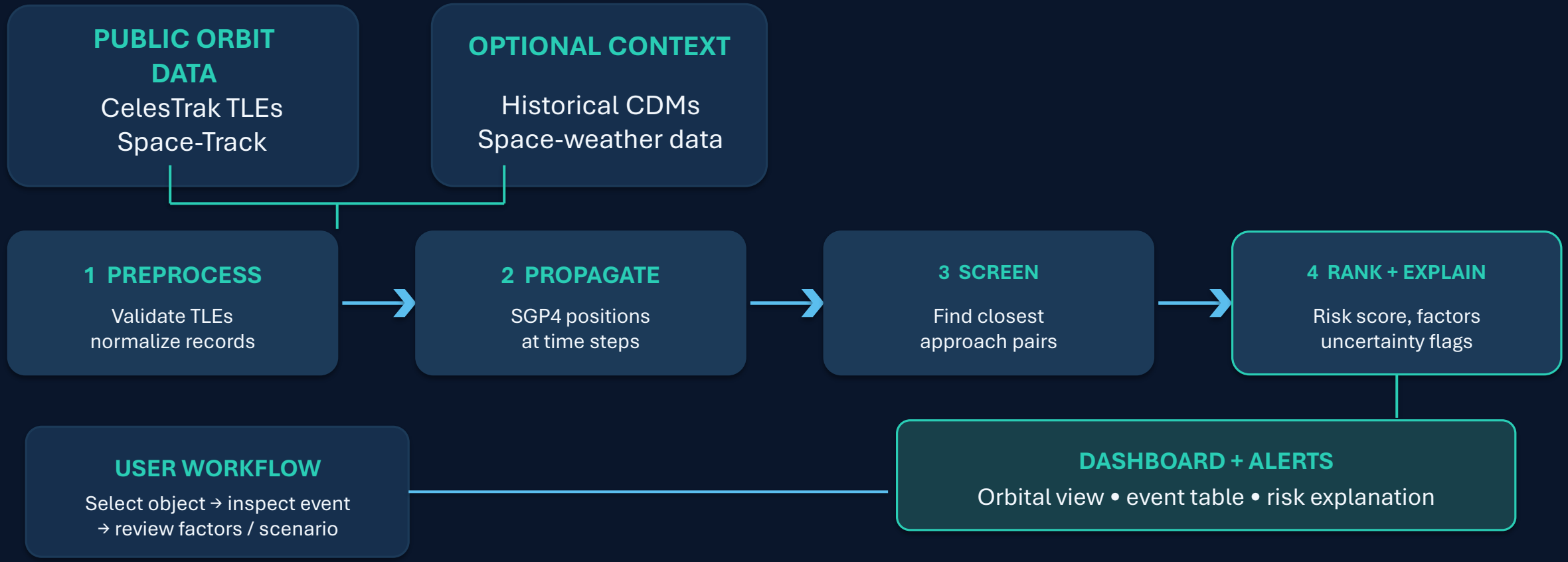
Instead of only identifying close approaches, our system ranks which events deserve attention first and explains the factors driving each risk flag.

APPROACH	CURRENT LIMITATION	OUR DIFFERENCE
Tracking networks	Limited Coverage	Use available data efficiency
Orbit propagation	Prediction Uncertainity	Combine with AI risk analysis
Conjunction screening	Many events to analyze	Prioritize high risk events
AI risk models	Can be difficult to explain	Explainable risk scoring
Operator workflows	Time consuming	Simple automated dashboard.

CORE INNOVATION: Existing systems track and predict → Our system helps prioritize and understand the risks.

System architecture & workflow

A lightweight batch pipeline: public data in, ranked conjunction events and visual explanations out.



Technology stack & implementation plan

Choose familiar, low-cost tools; keep the prototype modular and achievable.

PROPOSED STACK

DATA	CelesTrak public TLEs; optional Space-Track / CDMs
ORBIT PROPOGATION	Python • SGP4 / Skyfield • NumPy
RISK ANALYSIS	Pandas • closest-approach screening • explainable score
AI/ML	Scikit-learn if suitable labelled data is available
DASHBOARD AND STORAGE	Streamlit • Plotly • local CSV/SQLite storage
WHY THIS STACK?	Open-source • Low-cost • Lightweight • Fast to prototype • No GPU required for the core system

BUILD PLAN

1	**DATA + SETUP**	Load a bounded TLE dataset and validate object IDs, timestamps and data quality.
2	**ORBIT ENGINE**	Propagate objects using SGP4 and verify predicted positions.
3	**CONJUNCTION SCREEN**	Calculate closest approaches and generate candidate events.
4	**RISK + UI**	Apply explainable risk scoring and display event factors, rankings and alerts.
5	**TEST + DEMO**	Validate against reference/synthetic scenarios, measure performance and demonstrate one complete risk scenario.

TEAM RESPONSIBILITIES

M1: Data & Orbit
M2: Conjunction Screening
M3: Risk & AI
M4: Dashboard & Visualization
M5: Integration, Testing & Documentation

Expected outcomes & demo plan

A clear, testable demonstration with explicit boundaries and success measures.

PLANNED DELIVERABLES

- Working prototype using a bounded public TLE dataset
- Orbit propagation and close-approach event table
- Explainable screening risk score (not certified Pc)
- Interactive dashboard with trajectory / event view
- Short evaluation report and reproducible demo scenario

LIVE DEMO FLOW

1. **Load sample**
   Fetch/import orbital elements
2. **Propagate**
   Visualize future object positions
3. **Screen**
   Identify a close approach
4. **Explain**
   Show score, drivers and data caveat
5. **Review**
   Inspect event and optional what-if case

MEASURES OF SUCCESS

- Position error vs reference data
- Conjunction detection: precision / recall
- False-alert rate on test scenarios
- Processing time for chosen object set

RISKS & DEPENDENCIES

- TLE quality / age limits accuracy
- Historical labels / covariance may be unavailable
- Use synthetic cases for pipeline testing; label them clearly
- No automatic maneuver commands